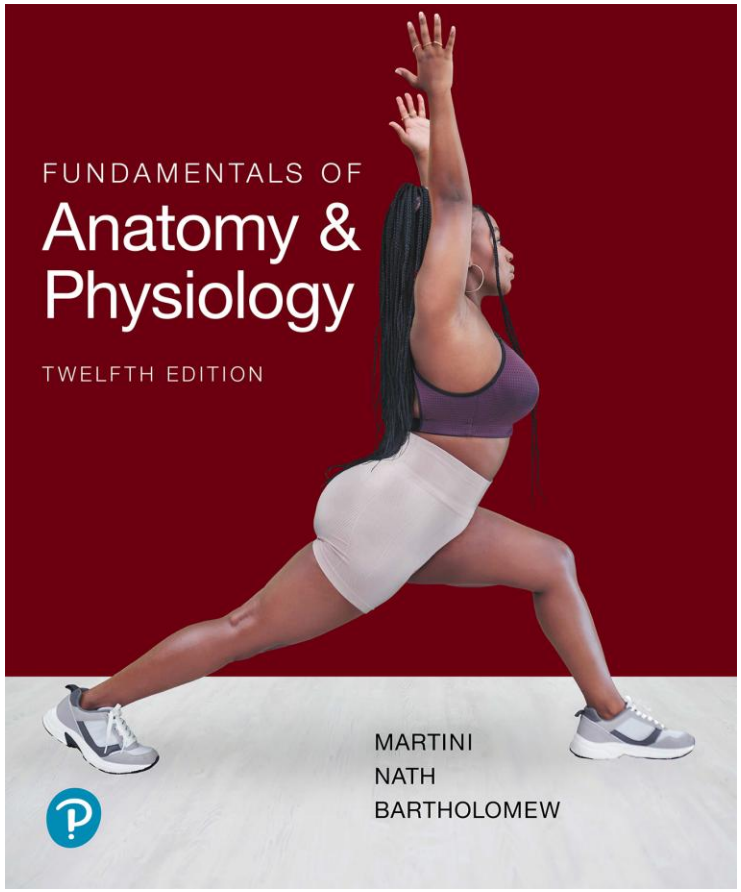
Fundamentals of Anatomy & Physiology

Twelfth Edition



Chapter 14

The Brain and Cranial Nerves

Lecture Presentation by Dr. Virginia Irintcheva, Truckee Meadows Community College

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Learning Outcomes (1 of 2)

14-1 Name the major brain regions, vesicles, and ventricles, and describe the locations and functions of each.

14-2 Explain how the brain is protected and supported, and discuss the formation, circulation, and function of cerebrospinal fluid.

14-3 Describe the anatomical differences between the medulla oblongata and the spinal cord and identify the main components and functions of the medulla oblongata.

14-4 List the main components of the pons and specify the functions of each.

14-5 List the main components of the midbrain and specify the functions of each.

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Introduction to the Brain and Cranial Nerves

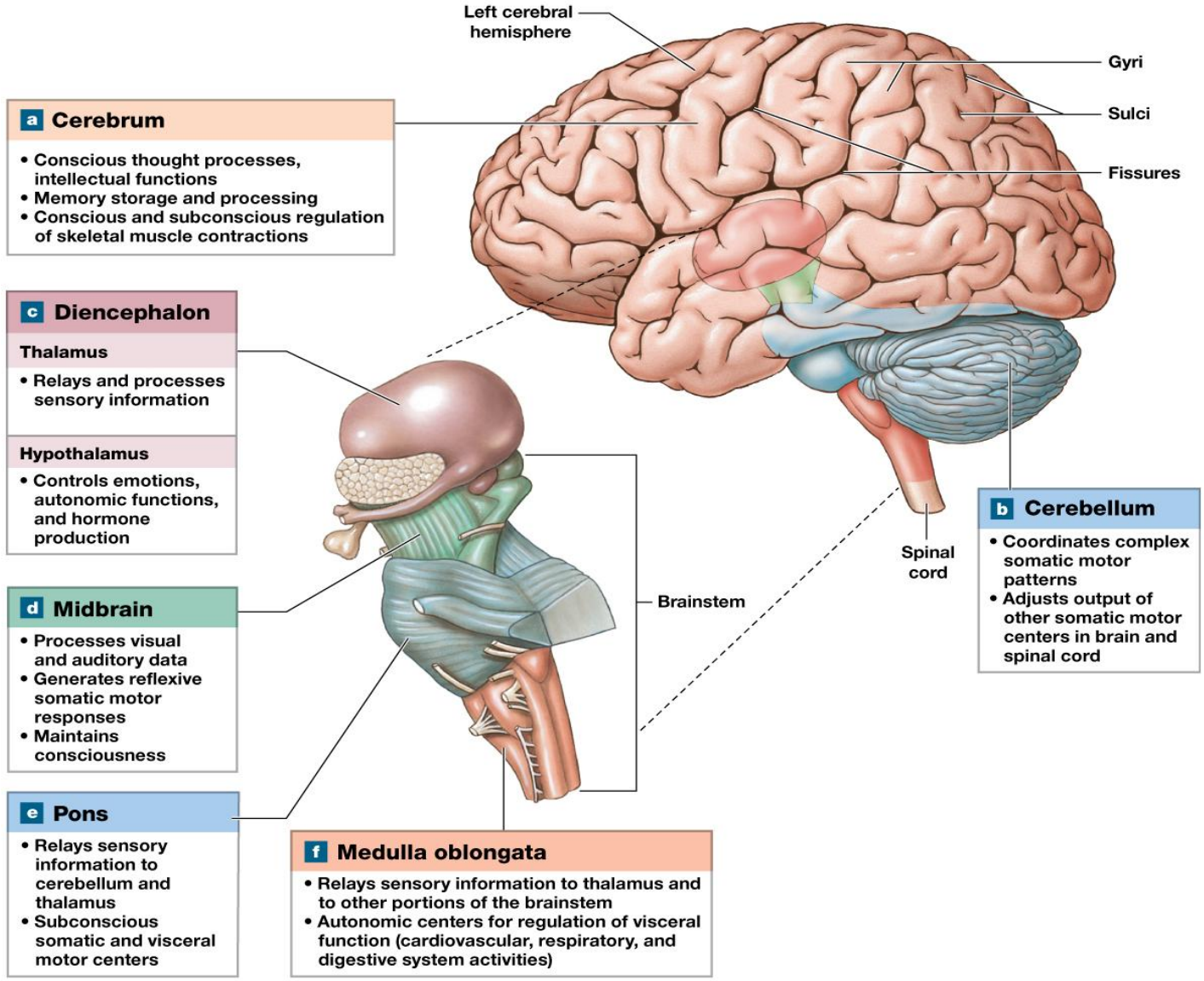
- The adult human **brain** contains almost 97 percent of body's nervous tissue
  - Average weight is 1.4 kg (3 lb)
  - "Typical" volume is 1200 mL
    - Ranges from 750 to 2100 mL in functionally normal individuals
  - Brains of males are about 10 percent larger than those of females
    - No correlation between brain size and intelligence

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Figure 14–1

An Introduction to Brain Structures and Functions



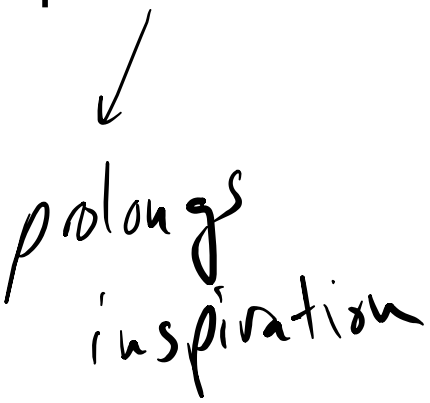
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14-4 The Pons (1 of 2)

Learning Outcome: List the main components of the pons and specify the functions of each.

- The **pons** contains:
  - Sensory and motor nuclei of cranial nerves (V, VI, VII, VIII)
  - Nuclei involved with the control of respiration
    - Apneustic center and pneumotaxic center





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